

Beyond Black-Scholes: Fitting Lévy Processes to Stock Returns

Abstract

In this project I fit five distributional models to 6,287 daily S&P 500 log-returns from January 2000 to December 2024: the Gaussian benchmark that underlies Black-Scholes, the Laplace (my inter-mediary distribution for Variance-Gamma), the Student-t, and two Lévy models built by Brownian subordination, the Variance-Gamma (Madan, Carr and Chang, 1998) and the Normal Inverse Gaussian. I estimated everything twice, by maximum likelihood and again in a Bayesian setting with the No-U-Turn Sampler. This meant the headline parameters come with full posteriors rather than point estimates.

The fitted Student-t degrees of freedom are 2.648, with a 94% credible interval of (2.46, 2.87) sitting entirely above the variance singularity at 2: the tails are heavy, and the variance is finite.

At the 99% level the Gaussian understates Expected Shortfall by 79.5% against the Student-t and using the Gaussian number to set capital against the NIG, the only model that survived every goodness-of-fit and posterior predictive check I ran, leaves a 37% shortfall.

A rolling out-of-sample backtest of one-day Value-at-Risk over 5,787 days, scored with the Christoffersen (1998) conditional coverage framework, confirms both halves of the story: the Gaussian produces 2.6 times the nominal number of 99% violations while the NIG cuts the excess to 1.6, yet every model, at every level, fails the independence test, because no static distribution can time the volatility clustering documented by Cont (2001).

A closing analysis reads the four crisis episodes of the sample through the Black Swan lens of Taleb (2007) and finds the label is a property of the model, not the event: the worst day of the COVID-19 crash is a once-per- 10^{22} -years event under the Gaussian, once per 119 years under the NIG and once per decade under the Student-t, yet no iid model of any tail weight reproduces the way such days cluster: eleven or twelve 99% violations inside a single month is an arrangement independence puts below one in twenty thousand.

Choosing a heavier marginal therefore improves the measurement of tail risk by amounts that matter under the FRTB metrics (BCBS, 2013).

1. Introduction

Since the early 1970s, most of quantitative finance has run on one large assumption: daily returns are normally distributed. Black-Scholes is built on it, and much of classical risk management inherited it.

The problem is that the assumption fails in exactly the region a risk manager is paid to care about. Real daily returns have a sharper peak, far heavier tails, and a persistent negative skew: crashes hit harder and more often than the equivalent rallies.

Cont (2001) collected these regularities into a list of stylised facts that any serious model of returns has to reproduce and made a structural observation I use as a design rule throughout this project: matching the location, scale, tail decay and asymmetry of real returns takes at least four parameters. The Gaussian has two.

The question I set out to answer is practical. Regulators moved on from 99% Value-at-Risk to 97.5% Expected Shortfall as the market-risk capital metric under the Fundamental Review of the Trading Book (BCBS, 2013), and ES is a tail expectation, so it is far more sensitive to distributional shape than a quantile is.

If the Gaussian understates the tail, it understates capital. By how much? And do the Lévy alternatives, which were built precisely to fix the tail, actually earn their extra parameters when tested properly?

My model set is a hierarchy of five. Gaussian is the two-parameter baseline. The Laplace, also two parameters, swaps thin tails for exponential ones and turns out to be the most informative cheap upgrade in the project. The student-t adds a third parameter that directly indexes tail heaviness.

The Variance-Gamma (VG) of Madan, Carr and Chang (1998) and the Normal Inverse Gaussian (NIG) are the two four-parameter Lévy models, each built by running Brownian motion on a random clock, and each meeting Cont's four-parameter requirement exactly.

I tested this hierarchy in five ways, and the paper follows that sequence:

- First, in-sample fit: maximum likelihood on the full 25-year sample, judged by information criteria and goodness-of-fit tests (Section 5).
- Second, a full Bayesian re-estimation with the No-U-Turn Sampler, which turns the important point estimates into posterior distributions and lets me make probability statements about the quantities that matter, such as whether the variance is finite (Sections 4.2 and 5.4).
- Third, checks on where the parameters move: across crises, across calendar structure, and against the VIX (Section 6).
- Fourth, and most demanding, an out-of-sample rolling backtest in which every model had to forecast tomorrow's VaR every day for 23 years and have the

forecasts scored with the Christoffersen (1998) conditional coverage tests (Section 7).

- Fifth, a reading of the four crisis episodes against Taleb's (2007) Black Swan criteria, which uses the fitted models to answer a question people usually argue about in words: how surprising was each crisis, with hindsight and at the time (Section 8).

The findings are compressed into four sentences. The Gaussian fails where failure is most expensive, understating 99% ES by 79.5% in-sample and producing 2.6 times the nominal number of 99% violations out of sample. Most of the improvement over the Gaussian comes from simply allowing exponential tails, but the NIG's extra structure is real: it is the only model that passed every distributional test I ran, in-sample and posterior predictive alike, and it came closest to nominal coverage out of sample.

No static distribution, however well shaped, passed the independence test, because violations arrive in bursts that only a dynamic volatility model could anticipate. And the Black Swan itself turns out to be model-relative: the crises that are impossible under the Gaussian are once-a-decade to once-a-century events under the heavy tails, while their timing stays outside every model in the class.

2. Data

The dataset is 6,287 daily log-returns on the S&P 500 index, $r_t = \log(S_t / S_{t-1})$, from January 2000 to December 2024. I chose the 25-year window deliberately: it contains four distinct stress episodes, the dot-com crash (March 2000 to October 2002), the Global Financial Crisis (October 2007 to March 2009), the COVID-19 shock (February to June 2020) and the Fed rate-hike cycle (2022 to 2023). These four windows are shaded consistently in every figure and reappear throughout the analysis, because it turns out they are not four examples of the same thing.

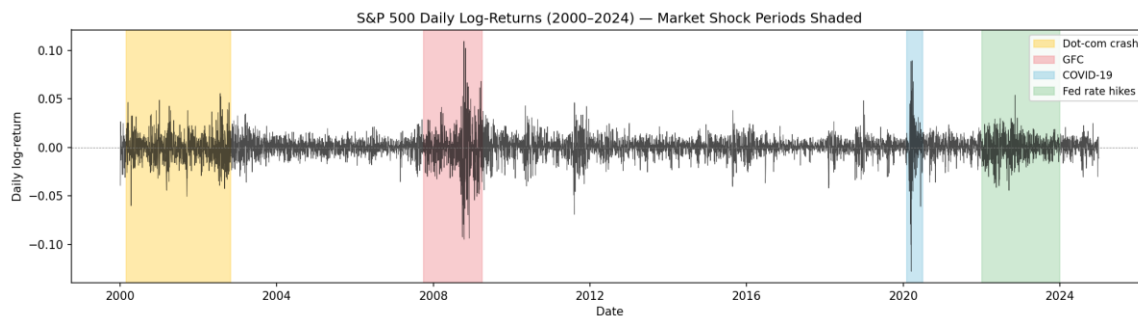


Figure 1. S&P 500 daily log-returns (2000 to 2024) with the four market shock windows shaded.

Figure 1 shows the series. The unevenness is the first thing to notice: the calm of 2013 to 2019 barely looks like the same market as late 2008 or March 2020. The worst single day

in the sample is a -12.8% log-return on 16 March 2020. Under a Gaussian fitted to this data, a move of that size is a twelve-standard-deviation event, which is a polite way of saying the model considers it impossible; Section 8 puts a number on the impossibility (once per 5.6×10^{22} years, about four trillion ages of the universe). The sample skewness is -0.39 and the sample excess kurtosis is 10.4 , against 0 and 0 for a normal distribution.

Those numbers are the first two of Cont's (2001) stylised facts showing up on schedule: heavy tails and gain/loss asymmetry. The third, volatility clustering, is visible in Figure 1 as the banding of quiet and violent stretches, and it will matter enormously in Section 7. The fourth, aggregational Gaussianity, the tendency of returns to look more normal at longer horizons, appears in my weekly-frequency fits in Section 6.3.

The whole project can be read as a test of how much of this list a static distribution can absorb, and the answer is: everything except the clustering.

3. The models

3.1 The hierarchy

Every model in the project is a distribution for the daily log-return. I fit them as independent and identically distributed draws, which is a deliberate choice. It isolates the question of distributional *shape* from the question of *dynamics*, and Section 7 measures the price of it.

The Gaussian $N(\mu, \sigma^2)$ is the Black-Scholes benchmark and needs no introduction. Its tails decay like $\exp(-x^2)$, which is the root of every failure documented below.

The Student-t $t(v, \mu, \sigma)$ adds a degrees-of-freedom parameter v that directly controls tail decay: the density falls off like a power law, $|x|^{-(v+1)}$, and moments only exist up to order v .

This makes v itself the object of interest. If the fitted v lands below 4, kurtosis is infinite; below 3, skewness is undefined; below 2, even the variance diverges. Where v falls relative to those thresholds is an empirical statement about how wild the market actually is.

The Variance-Gamma process of Madan, Carr and Chang (1998) is built by subordination: Brownian motion with drift θ and volatility σ is evaluated not at calendar time but at a random clock $G(t)$, a Gamma process with unit mean rate and variance rate v .

Economically the clock is market activity: it runs fast in busy periods and slow in quiet ones. The resulting log-return distribution $VG(\mu, \sigma, \theta, v)$ has four parameters with clean jobs: μ locates, σ scales, θ signs the skew (negative for equity-style left skew), and v sets the tail weight through the randomness of the clock.

The VG is a pure-jump process of finite variation with an infinite arrival rate of small jumps, which Madan, Carr and Chang (1998) argue is exactly what lets it behave diffusively at fine scales while still producing occasional large moves. Its density involves a modified Bessel function of the second kind, K_λ , which shapes all the numerical work in Section 4.

The Laplace distribution earns its place in the hierarchy through the VG. Set $\theta = 0$ and $\nu = 1$ and the Gamma clock becomes an Exponential; the symmetric normal variance-mean mixture then collapses to a pair of back-to-back exponentials, which is the Laplace $L(\mu, b)$ with density $\exp(-|x-\mu|/b)/(2b)$. It has the same parameter count as the Gaussian, exponential rather than Gaussian tails, an excess kurtosis of exactly 3, and a closed-form MLE (μ is the sample median, b the mean absolute deviation). It is the cheapest possible test of the proposition that the tails, not anything more exotic, are what the Gaussian gets wrong. From Section 5.4 onward the Laplace also stands in for the VG family in the Bayesian and backtesting work, a representation choice I return to in the limitations.

The Normal Inverse Gaussian $NIG(\mu, \alpha, \beta, \delta)$ is the second subordinated model: the same construction with an Inverse Gaussian clock instead of a Gamma one. Its density is

$$f(x) = \alpha\delta \cdot \exp(\delta\gamma + \beta(x-\mu)) \cdot K_1(\alpha \cdot q(x)) / (\pi \cdot q(x)), \text{ where } q(x) = \sqrt{(\delta^2 + (x-\mu)^2)} \text{ and } \gamma = \sqrt{(\alpha^2 - \beta^2)}.$$

The parameters again divide the labour: α controls tail decay (larger α means lighter tails), β the asymmetry subject to $|\beta| < \alpha$, δ the scale and μ the location.

3.2 What the hierarchy is for

The five models are ordered so that each comparison isolates one question. Gaussian against Laplace, at equal parameter count, asks whether the tail shape alone carries the improvement. Laplace against Student-t asks whether power-law tails beat exponential ones. Student-t against the four-parameter VG and NIG asks whether the asymmetry and the subordination structure add anything a symmetric heavy tail cannot. And VG against NIG asks whether the choice of random clock, Gamma against Inverse Gaussian, is detectable in 25 years of data. Every one of these questions gets a quantitative answer in Section 5.

4. Estimation

4.1 Maximum likelihood

The Gaussian and Laplace have closed-form MLEs. The Student-t is a routine three-parameter numerical optimisation.

I used the exponentially scaled Bessel function throughout: $\log K_{\lambda}(z) = \log kve(\lambda, z) - z$, which stays finite far into the tail. For the VG I optimised over $(\sigma, \theta, \nu, \mu)$ with L-BFGS-B from three starting points, keeping the best finite optimum.

The NIG has the awkward constraint $\alpha > |\beta|$, which I removed by reparametrising to $\xi = \beta/\alpha \in (-1, 1)$: box bounds on ξ then enforce the constraint automatically for any positive α , and the standard errors transform back to the natural parameters through the Jacobian.

All standard errors come from the observed Fisher information, computed with a central-difference Hessian whose step sizes scale with each parameter's magnitude.

Risk measures follow directly. For the Gaussian, Laplace and Student-t, VaR and ES at each level have closed forms; the Laplace ES is particularly tidy, sitting exactly one scale parameter b below the VaR because its exponential lower tail is memoryless. For the VG and NIG I computed VaR and ES by Monte Carlo from their mixture representations (a Gamma-Normal mixture for the VG, an Inverse-Gaussian-Normal mixture for the NIG) with one million draws, which makes the Monte Carlo error negligible at the levels I report.

4.2 Bayesian estimation

Maximum likelihood gives one best value per parameter and a curvature-based standard error. For the parameters this project actually cares about, that is not quite enough. The headline claim of Section 5.1 is that ν is near 2.6, and the interesting question is not the point estimate but how much posterior probability sits below the variance singularity at $\nu = 2$.

That is a question for a posterior, so I refitted four of the models (Gaussian, Laplace, Student-t and the full four-parameter NIG) in PyMC and sampled with the No-U-Turn Sampler.

The priors are weakly informative, anchored loosely on the MLE but widened well beyond the plausible range: $\mu \sim \text{Normal}(0, 0.001)$ for every model, $\text{HalfNormal}(0.02)$ for every scale parameter, $\nu \sim \text{Gamma}(2, 0.1)$ for the Student-t, and for the NIG a reparametrisation to (γ, β) with $\gamma \sim \text{Gamma}(4, 4/52)$ and $\beta \sim \text{Normal}(0, 15)$, recovering $\alpha = \sqrt{\gamma^2 + \beta^2}$ so the $\alpha > |\beta|$ constraint holds by construction.

The ν prior deserves a sentence, because it embodies the one methodological decision I thought hardest about. It would have been easy to restrict $\nu > 2$ and guarantee a finite variance.

I did not, because that restriction answers my main question before the data gets a vote. The prior I used has mean 20 and if anything leans towards light tails; any pull down towards $\nu \approx 2.6$ has to come from the likelihood.

I accepted each prior only after a prior predictive check: simulate return histories from the prior alone and confirm they stay within plausible bounds. The fraction of simulated

days beyond $\pm 25\%$, a move larger than anything in the sample, was between 0.0001% and 0.08% across the four models. Had any prior produced absurd histories I would have tightened it; none did.

The NIG was the real implementation work. Its log-density contains K_1 , and NUTS needs gradients through it. PyTensor's Bessel implementation carries correct analytic derivatives, so I wrote the NIG log-density directly as a custom distribution using the same scaled-Bessel trick as the MLE code, checked it against an independent reference implementation (the two agreed to fifteen decimal places), and let the sampler differentiate through it.

Convergence was clean everywhere: four chains per model, split $\hat{R}$ of 1.00 on every parameter, bulk effective sample sizes above 2,000, tail effective sample sizes above 2,100 (the tail figure is the one that matters here, since tail quantiles are what I report), and Monte Carlo standard errors small relative to posterior spread. Figure 5 in Section 5.4 shows the NIG traces, the hardest of the four; the chains sit on top of each other with no drift.

5. In-sample results

5.1 Parameter estimates

Table 1. Full-sample MLEs, 6,287 daily returns. Standard errors in parentheses.

Model	Parameters (SE)	Log-lik
Gaussian	$\mu = +0.000223$ (0.000154), $\sigma = 0.012234$ (0.000109)	18,764.3
Laplace	$\mu = +0.000602$ (0.000102), $b = 0.008078$ (0.000102)	19,649.9
Student-t	$\nu = 2.648$ (0.110), $\mu =$ $+0.000656$ (0.000110), $\sigma =$ 0.007077 (0.000128)	19,666.7
VG	$\sigma = 0.011629$ (0.000163), $\theta =$ -0.000660 (0.000149), $\nu =$ 1.174 (0.043), $\mu =$ $+0.000883$ (0.000023)	19,665.5
NIG	$\alpha = 52.34$ (2.79), $\beta = -6.10$ (1.50), $\delta = 0.007574$ (0.000217), $\mu = +0.001111$ (0.000150)	19,691.7

Three estimates carry the story. The Student-t ν of 2.648 sits below 4 and below 3: the data is consistent with infinite kurtosis and undefined skewness, though the 95% confidence interval (2.43, 2.87) keeps the variance finite. The VG θ and NIG β are both negative, the left skew of Cont's (2001) gain/loss asymmetry showing up as fitted

parameters. And every non-Gaussian log-likelihood beats the Gaussian's by around nine hundred units, an almost absurd margin for one or two extra parameters.

One estimate deserves a cross-reference. Madan, Carr and Chang (1998) fitted the VG to S&P 500 data from 1992 to 1994 and found the statistical process nearly symmetric with modest kurtosis. My 2000 to 2024 sample gives a VG v of 1.17 against their 0.002 in comparable units, and a clearly negative θ . Twenty-five years containing four crises is a very different market from the placid early 1990s, which is rather the point of the sample.

5.2 Which model fits

Table 2. Model comparison. Δ AIC is against the Gaussian; lower is better. KS is Kolmogorov-Smirnov on the full sample.

Model	k	AIC	BIC	Δ AIC	KS D	KS p
Gaussian	2	-37,524.5	-37,511.0	0	0.0938	<0.001
Laplace	2	-39,295.7	-39,282.2	-1,771.2	0.0180	0.034
Student-t	3	-39,327.5	-39,307.2	-1,802.9	0.0182	0.030
VG	4	-39,322.9	-39,296.0	-1,798.4	0.0129	0.250
NIG	4	-39,375.3	-39,348.4	-1,850.8	0.0113	0.399

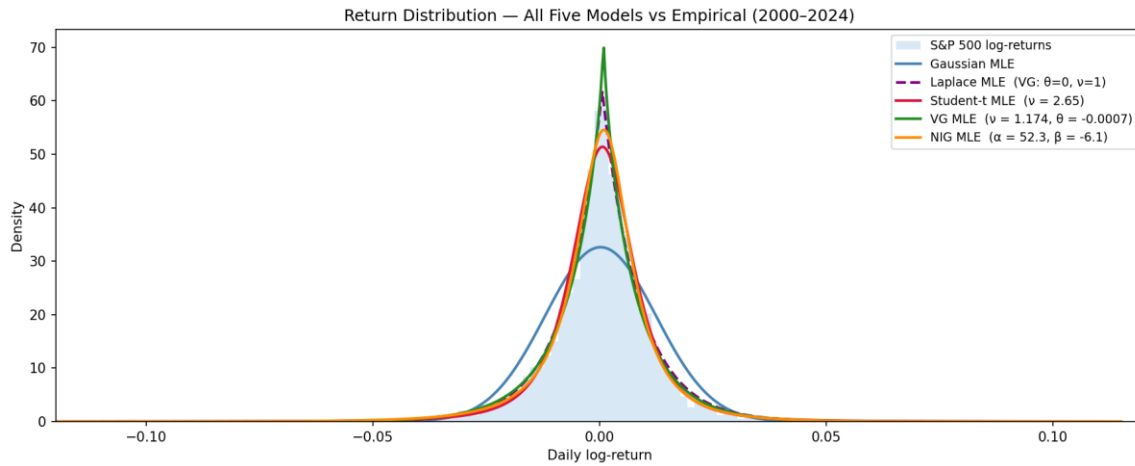


Figure 2. The return histogram with all five fitted PDFs. The four non-Gaussian curves are nearly indistinguishable at this scale; the Gaussian is visibly too flat at the peak and too fat in the shoulders.

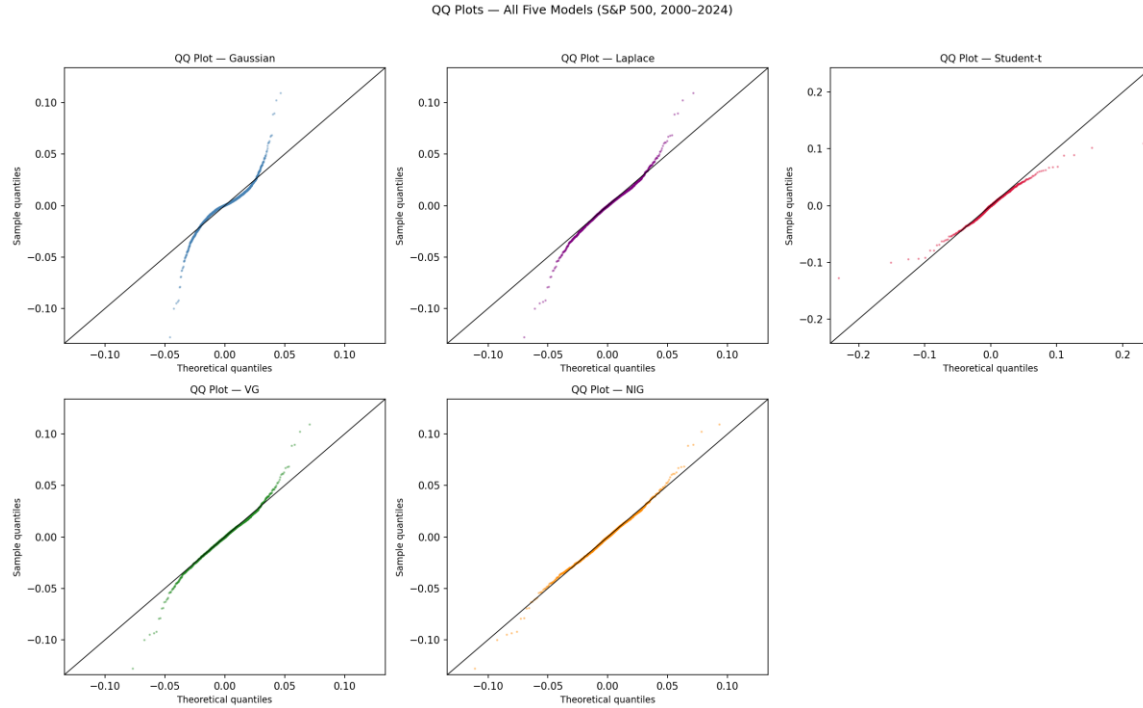


Figure 3. QQ plots. The Gaussian S-shape (top left) is the signature of heavy tails; the NIG sits closest to the diagonal throughout.

The answers to the four hierarchy questions of Section 3.2 are from Table 2. Tail shape alone is worth $\Delta\text{AIC} = -1,771$: the Laplace, with the Gaussian's parameter count, captures 96% of the best model's total gain

The full VG is actually slightly worse than the Student-t on AIC. However, the NIG beats everything, by 26 log-likelihood units over the VG on identical parameter counts, so the choice of random clock is detectable: the Inverse Gaussian subordinator captures something in this data that the Gamma one cannot.

On the KS test only the two Lévy models survive; the NIG ($D = 0.0113$, $p = 0.40$) fits best of all.

5.3 What it costs to be wrong

Table 3. Daily VaR and ES under each fitted model. Negative values are losses. 97.5% is the FRTB ES level.

Level	Gaussian	Laplace	Student-t	VG	NIG
VaR 95%	−1.99%	−1.80%	−1.69%	−1.92%	−1.88%
ES 95%	−2.50%	−2.61%	−3.00%	−2.82%	−3.08%
VaR 97.5%	−2.38%	−2.36%	−2.37%	−2.54%	−2.65%
ES 97.5%	−2.84%	−3.17%	−4.02%	−3.45%	−3.94%
VaR 99%	−2.82%	−3.10%	−3.52%	−3.36%	−3.77%
ES 99%	−3.24%	−3.91%	−5.81%	−4.28%	−5.19%

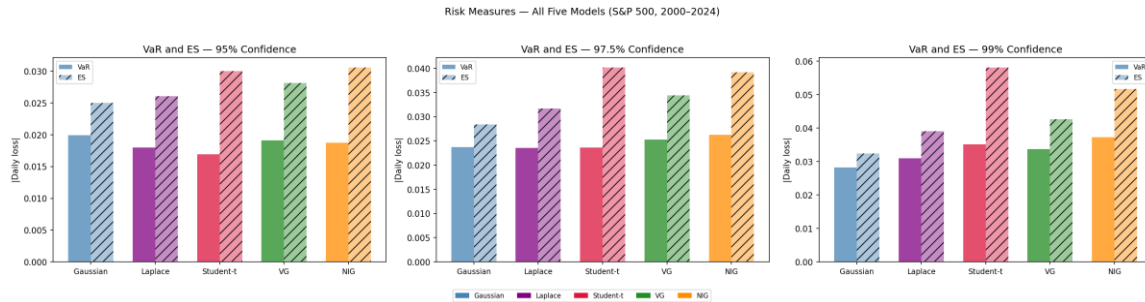


Figure 4. VaR and ES at 95%, 97.5% (the FRTB level) and 99%. The models agree closely on VaR at moderate levels and diverge enormously on ES in the deep tail.

Table 3 has two patterns worth pulling out:

- First, the models nearly agree on 95% and 97.5% VaR, the quantities everyone used to report, and then diverge violently on ES: at 99%, the span runs from -3.24% (Gaussian) to -5.81% (Student-t), a 79.5% gap. The regulatory shift from VaR to ES (BCBS, 2013) is exactly the shift from the metric these models agree on to the metric they disagree on, which is why the distributional choice suddenly has capital consequences.
- Second, the shortfall is already material at the level the regulation actually uses: at 97.5%, the Gaussian ES of -2.84% sits 28% below the NIG's -3.94% . A desk that sets ES-based capital from the Gaussian holds roughly 37% too little against the NIG benchmark at 99%, and the deficit does not wait for the deep tail to appear.

There is one instructive oddity in the VaR rows: at 95%, the Gaussian VaR is the most conservative of the five. The Student-t, having committed its probability mass to the far tail, has less left in the shoulders, so its 5% quantile sits closer to zero.

5.4 The Bayesian answers

Table 4. Posterior summaries against the MLE. Intervals are 94% highest-density intervals.

Model	Parameter	Posterior mean	94% HDI	MLE
Gaussian	μ	+0.000215	(−0.00007, +0.00049)	+0.000223
	σ	0.012233	(0.01202, 0.01244)	0.012234
Laplace	μ	+0.000599	(+0.00040, +0.00080)	+0.000602
	b	0.008081	(0.00790, 0.00827)	0.008078
Student-t	μ	+0.000647	(+0.00043, +0.00084)	+0.000656
	σ	0.007084	(0.00686, 0.00733)	0.007077
	ν	2.658	(2.46, 2.87)	2.648
NIG	μ	+0.001080	(+0.00081, +0.00135)	+0.001111
	α	52.35	(47.1, 57.7)	52.34
	β	−5.86	(−8.75, −3.25)	−6.10
	δ	0.007575	(0.00717, 0.00798)	0.007574

NUTS trace — NIG

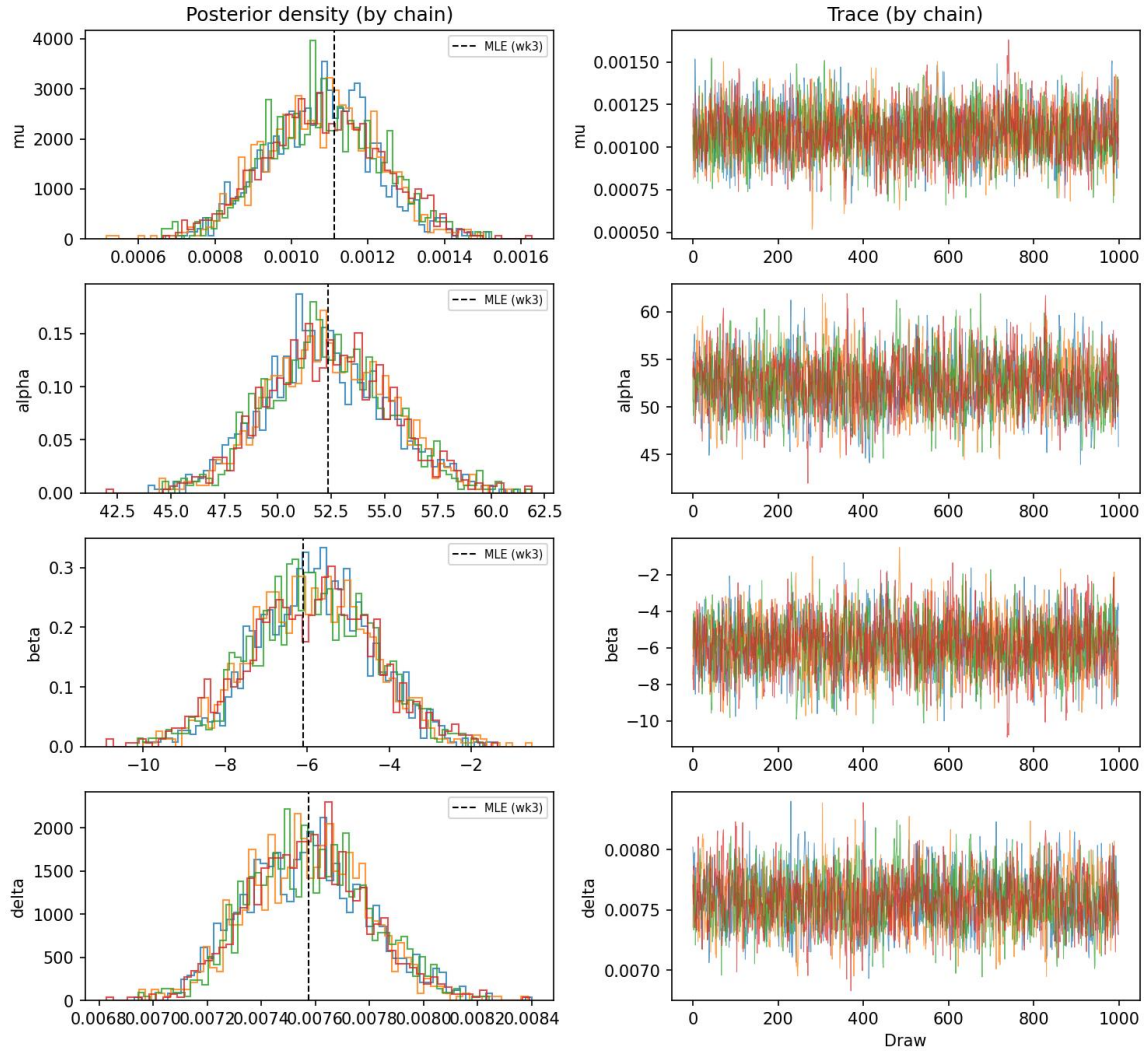


Figure 5. NUTS diagnostics for the NIG, the hardest model of the four: posterior densities per chain against the MLE (left) and the raw chains (right). Four parameters and a Bessel function in the density, and the mixing is as clean as the Gaussian's.

With 6,287 observations and loose priors, the posteriors reproduce the MLEs almost exactly, which is reassurance about both rather than news. The two results that justify the Bayesian effort are the intervals. The entire 94% interval for ν , (2.46, 2.87), sits above 2: the finite variance is not a point estimate that happens to land north of the singularity, it holds across the whole credible range, under a prior that leaned the other way. And the entire interval for the NIG's β , (−8.75, −3.25), sits below zero: the left skew is a stable feature of the data, not something that could round away.

5.5 Posterior predictive checks

For each model I drew 1,000 parameter values from the posterior, simulated a full 6,287-day replicate history from each, and scored the real data against the replicate cloud with two-sided posterior predictive p-values: a p near 0.5 means the model produces data like ours comfortably, a p near 0 or 1 means the real data sits outside almost every replicate.

Table 5. Posterior predictive p-values. Entries marked † fall outside (0.05, 0.95). ACF²(1) is the lag-1 autocorrelation of squared returns.

Statistic	Gaussian	Laplace	Student-t	NIG
Std deviation	0.99 †	0.00 †	0.05	0.82
Excess kurtosis	0.00 †	0.00 †	0.00 †	0.21
Skew	0.00 †	0.00 †	0.81	0.56
Minimum	0.00 †	0.00 †	0.22	0.36
Maximum	0.00 †	0.01 †	0.05	0.28
ACF ² (1)	0.00 †	0.00 †	0.00 †	0.00 †
VaR 95%	0.01 †	0.13	0.00 †	0.93
ES 95%	0.00 †	0.00 †	0.92	0.55
VaR 97.5%	0.00 †	0.03 †	0.08	0.37
ES 97.5%	0.00 †	0.00 †	0.52	0.50
VaR 99%	0.00 †	0.00 †	0.94	0.16
ES 99%	0.00 †	0.00 †	0.23	0.73

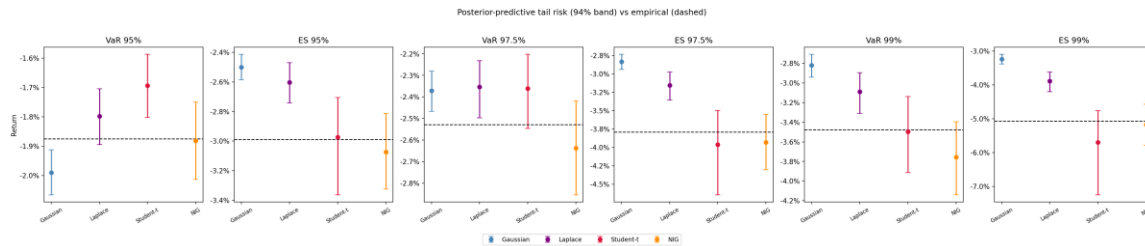


Figure 6. Posterior predictive VaR and ES at the three levels. Each marker is a model's replicate median with its 94% band; dashed lines are the observed values. The Gaussian bands sit clear of the truth beyond 95%; the NIG covers it on all six statistics.

The table is the project's in-sample verdict in one place. The Gaussian fails everything beyond the centre: across a thousand replicate histories its worst simulated day never got past about -5% , against the real -12.8% .

The Laplace closes roughly half the tail gap and then stops at exactly the excess kurtosis of 3. The Student-t captures the deep tails, but with ν near 2.66 its higher moments do not exist, so its moment-based checks, including the skew pass, carry no information. The NIG passes all eleven marginal checks with tight, meaningful bands, including the skew (replicates centred at -0.54 against the observed -0.39) and the FRTB 97.5% ES.

All four models fail the clustering check identically (observed 0.32 against bands on zero), because no iid model can cluster, and that shared failure is what motivates the out-of-sample test in Section 7.

6. Parameters that will not sit still

Everything in Section 5 assumed one distribution for 25 years. This section tests that assumption from three directions and finds it wrong in interesting, structured ways.

6.1 Two kinds of crisis

Fitting each model separately inside the four shock windows splits the crises into two families.

Table 6. Sub-period estimates, each window fitted independently.

Period	n	Gaussian σ (ann.)	Student-t ν	VG ν	VG θ	NIG α	NIG β
Full sample	6,287	19.4%	2.65	1.17	−0.0007	52.3	−6.1
Dot-com crash	671	23.5%	6.53	0.40	+0.0019	98.6	+10.0
GFC	378	38.3%	2.61	1.25	−0.0030	25.9	−2.7
COVID-19	104	50.5%	2.29	1.91	−0.0050	17.8	−4.2
Fed rate hikes	501	19.5%	6.53	0.49	−0.0000	111.9	−3.9

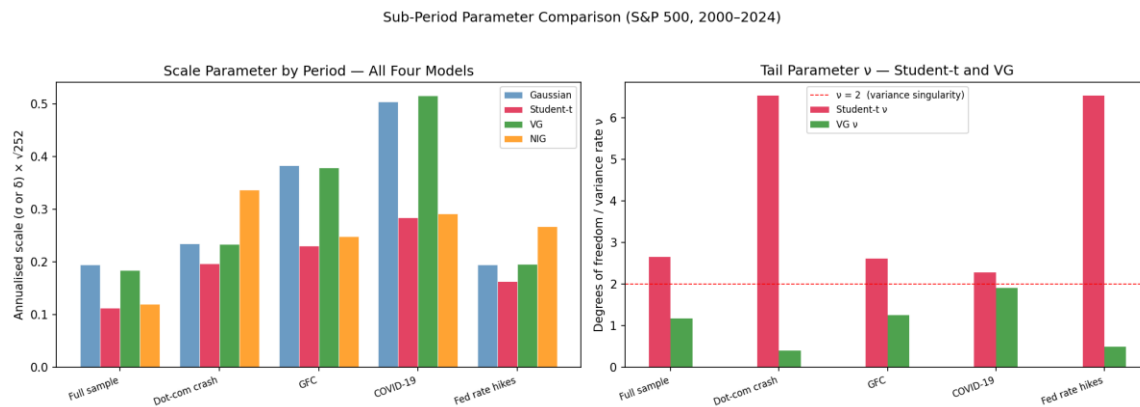


Figure 7. Annualised scale by period and model (left) and the tail parameters (right); the dashed line is the $\nu = 2$ variance singularity.

The GFC and COVID windows return ν near 2.3 to 2.6 and NIG α of 18 to 26: clusters of extreme single days, tails at their heaviest exactly when volatility is also at its highest. The dot-com and Fed-hike windows were high-volatility but nearly Gaussian in shape (ν

near 6.5, α near 100). The Lévy models' AIC edge over the Gaussian grows from 15 to 23 units in calm markets to 73 to 79 inside the GFC: the extra parameters pay off precisely in the worst regimes.

6.2 Tracking the parameters through time

Year-by-year VG and NIG fits (with μ pinned to zero, since 250 observations cannot identify a daily drift) turn the parameters into time series.

The scale parameters trace the crises, the VG's annualised σ peaking at 39.8% in 2008 and 31.3% in 2020 against 6.5% in 2017. Tail heaviness is itself time-varying: NIG α bottoms out near 22 in exactly 2008 and 2020, and the most negative skew readings land on the dot-com unwind, the GFC and the 2022 drawdown.

In the calmest years the NIG runs into its Gaussian limit, where α and δ grow together essentially unbounded: each carries a huge standard error and only their product is identified. That ridge recurs throughout the project: roughly a third of quarterly fits sit on it, and it later broke a library quantile routine in the backtest (Section 7.1).

At quarterly frequency there is enough data to regress the parameters on the VIX. The NIG scale δ tracks it closely ($R^2 = 0.64$ in logs, $t = 9.7$ across the 67 well-identified quarters), and the skew parameters show a significant VIX relationship at quarterly frequency that annual fits lacked the power to find.

The VIX tells you how wide the distribution is.

6.3 The calendar

The calendar rounds out the picture of where the risk lives.

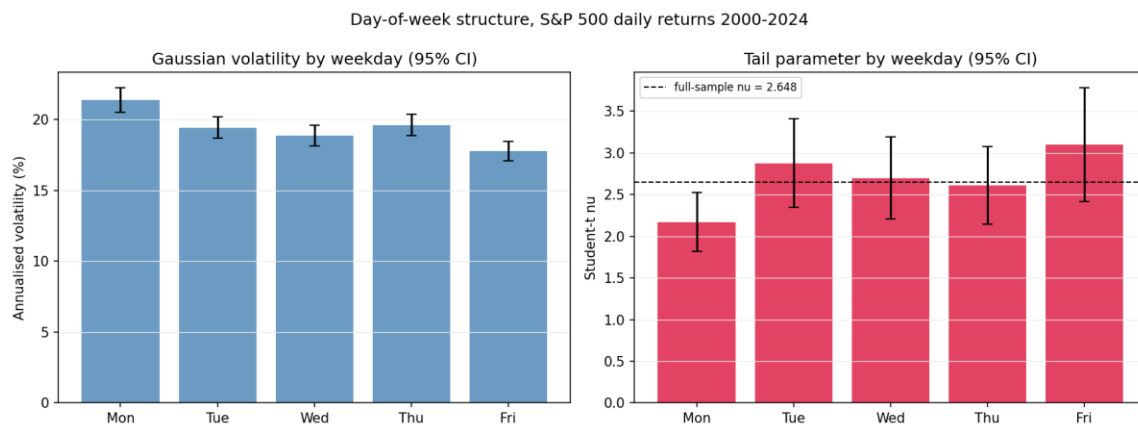


Figure 8. Annualised volatility by weekday with 95% intervals (left) and the per-day Student-t ν against the full-sample 2.648 (right).

The week has a shape, and it is a staircase. Monday, which carries the whole weekend inside it, is the most volatile day (21.4% annualised) and has much the heaviest tail ($v = 2.17$, brushing the variance boundary).

Friday is the calmest and lightest (17.8%, $v = 3.10$).

Likelihood-ratio tests settle the right resolution: a Monday / midweek / Friday grouping beats a pooled week decisively (LR = 42.9 on 4 df, $p < 0.0001$), and five separate days add nothing over the three groups ($p = 0.59$).

Second, splitting close-to-close returns at the open shows the closed market holds only about 20% of the variance but nearly all of the jumps: overnight $v = 2.14$ with excess kurtosis 36, against intraday $v = 2.79$ and kurtosis 5.

The index gaps on information that arrives while trading is shut, which is also what Monday's weekend tail is. Third, weekly returns fit with $v = 3.38$ against the daily 2.65: aggregation lightens the tail exactly as Cont's (2001) aggregational Gaussianity predicts, so every tail conclusion in this paper is specific to the daily horizon.

One null result is worth reporting because I expected the opposite: quarterly earnings seasons leave index-level volatility unchanged to the decimal (16.5%, 16.4%, 16.5% annualised before, during and after).

Single-name earnings surprises evidently diversify away in a 500-stock index; the tail risk that survives aggregation is macro.

7. The out-of-sample backtest

Everything so far has been in-sample. For the final test I made the models work for a living: each one refitted on a rolling window and asked, every day, for tomorrow's Value-at-Risk, with the forecasts scored by the realised returns they had never seen.

7.1 Design

I chose a rolling window of 500 trading days, about two years: long enough for a stable four-parameter NIG fit, short enough to adapt across regimes. I refitted every 21 trading days, roughly the monthly cycle a risk desk would run, and held parameters fixed between refits.

Each day the previous fit issued one-day-ahead VaR at 95%, 97.5% and 99%, taken as the corresponding quantile of the fitted distribution, and a violation is a realised return strictly below the forecast. That gives 5,787 out-of-sample days per model, January 2002 to December 2024, with 289.4, 144.7 and 57.9 violations expected at the three levels. The models are the Bayesian four, with the Laplace again standing in for the VG family.

One numerical fight worth recording: on calm windows the rolling NIG regularly lands on the Gaussian-limit ridge from Section 6.2, and the library quantile routine refused to converge there, so I solved the quantile myself, bracketing the root with the NIG's own

mean and standard deviation and running Brent's method on the CDF. On the ridge the quantile smoothly approaches the Gaussian one, which is the behaviour I wanted.

The scoring framework is Christoffersen (1998). His point is that the average violation rate on its own is not enough: a model can be right on average and still fail in bursts, understating risk exactly when it matters most. So he tests two things. The Kupiec test (LR_uc) asks whether violations happen at the promised rate of 5%, 2.5% or 1%. The independence test (LR_ind) asks whether a violation today makes one tomorrow more likely, when it should not. The joint test (LR_cc) simply requires both. A correct model produces the right number of violations, arriving at unpredictable times.

7.2 Coverage

Table 7. Rolling backtest over 5,787 out-of-sample days. Hits are observed violations; p-values from the chi-squared limits.

Model	Level	Expected	Hits	LR_uc	p_uc	LR_ind	p_ind	p_cc
Gaussian	95%	289	308	1.2	0.265	47.2	<0.0001	<0.0001
Gaussian	97.5%	145	223	37.4	<0.0001	39.7	<0.0001	<0.0001
Gaussian	99%	58	152	106.9	<0.0001	36.1	<0.0001	<0.0001
Laplace	95%	289	323	4.0	0.046	35.7	<0.0001	<0.0001
Laplace	97.5%	145	193	15.0	0.0001	34.6	<0.0001	<0.0001
Laplace	99%	58	110	37.5	<0.0001	28.0	<0.0001	<0.0001
Student-t	95%	289	358	16.0	0.0001	39.0	<0.0001	<0.0001
Student-t	97.5%	145	218	33.1	<0.0001	35.9	<0.0001	<0.0001
Student-t	99%	58	104	30.0	<0.0001	18.6	<0.0001	<0.0001
NIG	95%	289	313	2.0	0.159	37.5	<0.0001	<0.0001
NIG	97.5%	145	177	6.9	0.009	31.9	<0.0001	<0.0001
NIG	99%	58	91	16.3	0.0001	19.3	<0.0001	<0.0001

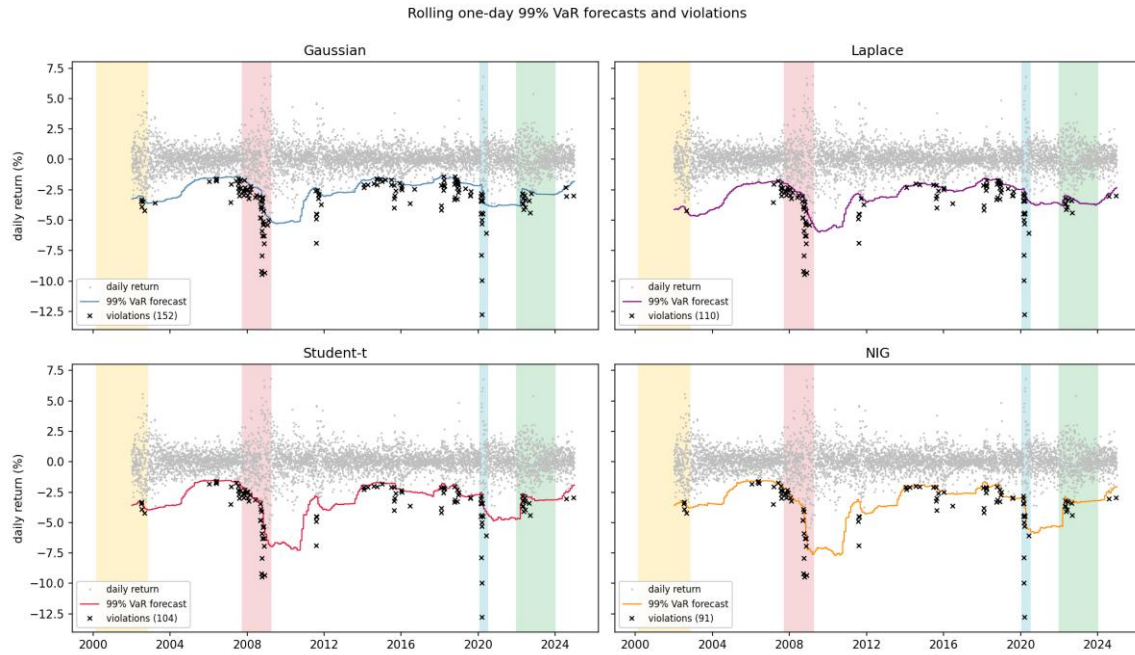


Figure 9. Rolling one-day 99% VaR forecasts against realised returns, violations marked. The crosses bunch inside the GFC and COVID windows for every model.

At 95%, the Gaussian passes unconditional coverage comfortably ($p = 0.27$) and the Student-t is the worst model in the table (358 hits against 289). This is the Section 5.3 geometry coming due: a tail heavy enough to cover crashes thins the shoulders where the everyday 5% quantile lives, so at moderate levels the heavy-tailed model under-covers while the thin-tailed one is fine. Tail heaviness is a budget, not a free improvement.

At 99% the ordering flips and the Gaussian collapses: 152 violations against 58 expected, a factor of 2.6, $LR_{uc} = 106.9$. This is the out-of-sample twin of the 79.5% in-sample ES gap. The NIG comes closest (91 hits, a factor of 1.57), then the Student-t and Laplace, and even the NIG is rejected on pure coverage: a static tail fitted to the last two years is still too short when the regime breaks.

7.3 Timing

The LR_{ind} column has no survivors: every model, at every level, is rejected with statistics between 18.6 and 47.2. The mechanism shows in the transition counts. After a Gaussian 95% violation, the chance of another violation the next day was about 16%, three times the unconditional rate. The models are not wrong at random times; they are wrong in runs.

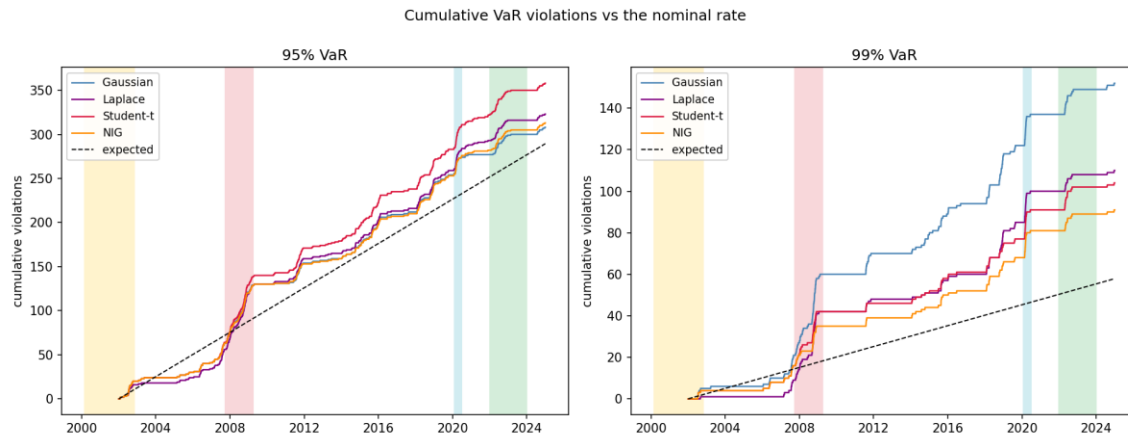


Figure 10. Cumulative violations at 95% (left) and 99% (right) against the nominal accrual (dashed). The staircases are flat for years and then vertical in late 2008 and March 2020.

Figure 10 is the independence failure and it is also the exact pattern Christoffersen (1998) documented for static interval forecasts in his original exchange-rate application: right on average, wrong in clusters. It is the out-of-sample confirmation of the posterior predictive ACF failure in Section 5.5, now with real forecasts and real money implications.

A rolling window softens the problem but cannot fix it: refitting monthly on two years of data always arrives about a month late to a regime change.

7.4 The regulatory view

Basel's backtesting regime keys the capital multiplier off the count of 99% violations in the trailing 250 days: green up to 4, yellow from 5 to 9, red at 10 or more, where the internal model is presumed broken.

Table 8. Share of out-of-sample days in each Basel zone (99% VaR, trailing 250 days), and the FRTB-style ES check at 97.5% (mean realised loss on breach days over the mean ES the model forecast for those days).

Model	Green	Yellow	Red	Worst 250-day count	ES shortfall ratio
Gaussian	53.2%	16.0%	30.8%	32	1.33
Laplace	63.8%	17.5%	18.7%	28	1.20
Student-t	65.3%	16.7%	17.9%	22	0.97
NIG	66.6%	19.3%	14.1%	15	1.07

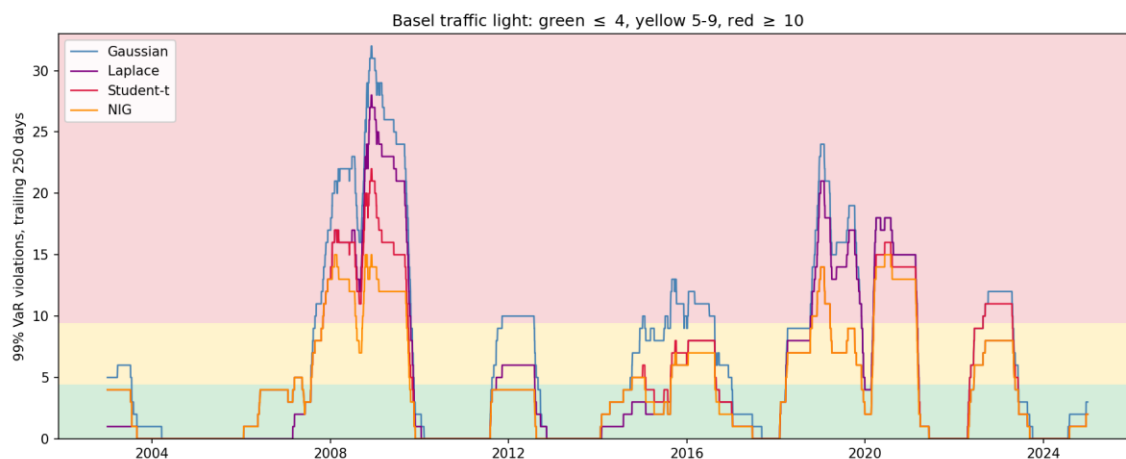


Figure 11. Trailing 250-day count of 99% violations per model over the Basel green, yellow and red zones.

A Gaussian desk spends almost a third of two decades in the red zone, and its worst 250-day window contains 32 violations, more than three times the red threshold.

The ES column says the same thing in FRTB units: on the days the Gaussian's 97.5% VaR was breached, the realised losses ran 33% past the ES the model had booked, and ES at 97.5% is precisely the number FRTB capital is built on (BCBS, 2013).

The Lévy tails roughly halve the red-zone time, the NIG's worst window is half the Gaussian's, and the NIG prices its own breaches to within 7%. The Student-t's promised shortfall is actually adequate (ratio 0.97), the mirror image of its 95% coverage failure: deep-tail honesty bought at the shoulders. But no model stays out of the red through 2008 and 2020, which is the traffic-light restatement of the independence failure.

8. Were the crises Black Swans?

Taleb (2007) gives a Black Swan three attributes.

It lies outside the realm of regular expectation, it carries an extreme impact, and human nature makes it explainable after the fact.

The second and third attributes belong to history and psychology.

The first one is statistics, and it hides the assumption this paper has been testing from the start: an event is only "outside regular expectation" relative to a model of what regular expectation is. This section prices each crisis under each fitted model. The unit is the implied return period: if a model assigns probability p to a daily log-return at or below the observed one, days that bad recur on average every $1/(252p)$ years under that model.

A two-year return period is a routine event; a thousand-year one puts the event effectively outside the model's world; anything beyond the age of the universe is the model declaring

the event impossible. The reference line throughout is 25 years, the length of the sample the events actually sit in.

I computed the probabilities from the closed-form CDFs where they exist and by numerical integration of the fitted density over the lower tail for the VG and NIG, reusing the scaled-Bessel machinery of Section 4.1.

8.1 Surprise with hindsight

Table 9. Implied return period in years of each crisis's worst single day under the five full-sample MLEs. The final column is the empirical recurrence of days at least that bad in the 6,287-day sample.

Crisis (worst day)	Gaussian	Laplace	Student-t	VG	NIG	Empirical
Dot-com, -6.00% (14 Apr 2000)	9,470	14.5	1.5	7.3	2.0	13 days, one per 1.9 y
GFC, - 9.47% (15 Oct 2008)	9.3×10^{11}	1,050	4.9	318	17.5	3 days, one per 8.3 y
COVID- 19, - 12.77% (16 Mar 2020)	5.6×10^{22}	62,400	10.6	11,400	119	1 day, one per 24.9 y
Fed hikes, -4.42% (13 Sep 2022)	28.1	2.0	0.7	1.3	0.7	33 days, one per 0.8 y

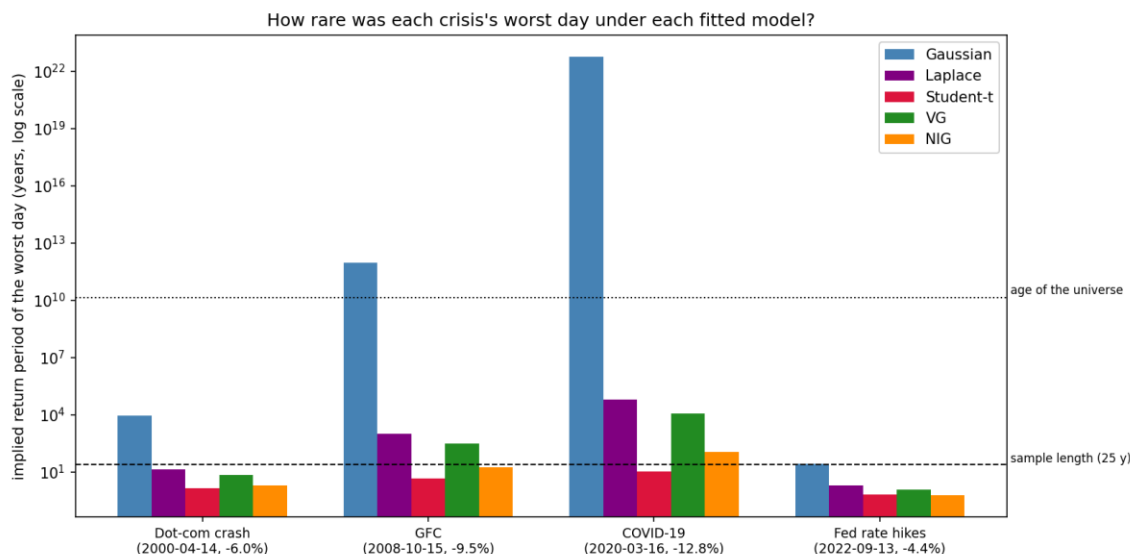


Figure 12. Implied return period (log scale) of each crisis's worst day under the five full-sample fits. The dashed line is the sample length; the dotted line is the age of the universe. The Gaussian's GFC and COVID-19 bars clear the age of the universe by factors of 67 and four trillion.

The empirical column is the referee. The Gaussian declares every jump event impossible: the GFC's worst day is a 67-ages-of-the-universe event, COVID-19's is four trillion ages, and even the mild -4.42% of September 2022, a day the sample serves up 33 times, gets a return period longer than the sample that contains it. Under the Gaussian all four crises are textbook Black Swans, and that is a statement about the Gaussian, not about the crises. The Student-t ($\nu = 2.648$) absolves everything: its implied recurrence matches the empirical column almost exactly at the two grind events and prices the COVID-19 day at once per decade, which inside a 25-year sample is not a swan of any colour. The NIG grades the same events one notch rarer, 17.5 and 119 years for the two jump days: grey swans in Taleb's own classification.

The Laplace and VG rows carry a warning for anyone who selects models by likelihood alone. These are the exponential-tailed models that Section 5.2 showed capturing nearly all of the AIC gain over the Gaussian, yet they price the COVID-19 day at once per 62,400 and 11,400 years, three to four orders of magnitude too rare.

The likelihood is dominated by the centre of the distribution, where they excel; the Black Swan question is decided ten standard deviations out, where they revert to Gaussian-style dismissal.

Only the power-law tail of the t, and at one remove the semi-heavy NIG tail, respects the extremes as well as the centre.

The Bayesian machinery of Section 5.4 says the verdict is not an estimation accident. Pushing 2,000 posterior draws per model through the same calculation for 16 March 2020, the Gaussian's 94% interval for the return period runs from $10^{21.9}$ to $10^{23.6}$ years, the Student-t's from 8 to 16 years, the NIG's from 65 to 230.

Within-model parameter uncertainty spans tenths of a decade; between-model disagreement spans 21 decades. Whether that day was a Black Swan is decided wholesale by the choice of tail, not by anything the data leave uncertain given the model.

8.2 Surprise at the time

Hindsight fits flatter every model, because the crisis being judged is inside the estimation sample pulling the tail outward. Taleb's actual criterion is prospective, so I refitted all five models on the 500 trading days (the backtest window of Section 7.1) ending the day before each crisis window opens, and scored the crisis under that frozen pre-crisis fit.

The dot-com episode drops out here: the sample starts in January 2000 and offers only 39 pre-crisis days.

Table 10. Crisis days under the frozen pre-crisis fits: return period of the worst day, mean log predictive density per crisis day, and the number of crisis days the model priced below one in a thousand.

Crisis	Model	RP of worst day (years)	Mean log-score	Days with $p < 0.001$
GFC (378 days)	Gaussian	4.2×10^{33}	-1.18	53
	Laplace	443,000	1.44	29
	Student-t	39.2	1.72	17
	VG	146,000	1.45	26
	NIG	1,820	1.62	20
COVID-19 (104 days)	Gaussian	4.5×10^{40}	-2.13	17
	Laplace	3.5×10^6	0.99	9
	Student-t	34.4	1.41	4
	VG	1.3×10^6	0.98	9
	NIG	2,100	1.22	5
Fed hikes (501 days)	Gaussian	1.19	2.91	0
	Laplace	0.84	2.98	0
	Student-t	0.36	2.95	0
	VG	0.41	2.95	0
	NIG	0.26	2.95	0

The Gaussian's surprise explodes out of sample. Trained on the calm of 2005 to 2007 it prices 15 October 2008 at once per 10^{33} years; trained on 2018 to 2019 it prices 16 March 2020 at once per 10^{40} . Its mean log-score is negative in both episodes, where a competent model of daily returns earns around +3. It assigned essentially zero probability to what happened, day after day, for months, and priced one GFC trading day in seven below one in a thousand.

The Student-t barely notices the loss of hindsight: 39 and 34 years against 4.9 and 10.6 with it. A power-law tail estimated on two calm years is still a power law, so the model keeps its scepticism about calm even when the recent data contain none.

At the moment it happened, the worst COVID-19 day was, under the t, a once-in-a-generation event: severe, but named and priced.

The NIG tells a more cautionary story: it loses three orders of magnitude out of sample (1,820 and 2,100 years against 17.5 and 119). The mechanism is the Gaussian-limit ridge of Sections 6.2 and 7.1: on calm two-year windows the NIG MLE drifts towards light tails, and hands the arriving crisis a far thinner tail than the full sample would. The t's tail survives calm training data; the NIG's does not. Under a regime where models are re-estimated on recent windows, which is how FRTB internal models actually operate (BCBS, 2013), that fragility is a capital-relevant property, and it does not show up in any full-sample ranking.

The 2022 tightening, finally, was no swan at all: every model including the Gaussian priced the whole episode within about a year's return period, and no model priced a single day of it below one in a thousand. A bear market delivered as five hundred ordinary down days is what every one of these distributions expects. And the lead-up behaviour separates the two jump crises on Taleb's "no warning" criterion: the GFC assembled itself in plain sight, with trailing volatility and drawdown elevated through 2007, while COVID-19 erupted from below-average volatility and a below-average VIX. COVID-19 is the closest thing in this sample to a genuine Black Swan, and even its magnitude sat within a once-in-a-generation tail under the right marginal.

8.3 Whole histories, and what remains

A stricter test asks each model to reproduce the sample as a whole. I simulated 4,000 independent 25-year histories (6,287 days each) from each full-sample fit and recorded, per history, the worst day, the worst 21-day stretch, and the number of days below three and five empirical standard deviations. The observed history has a worst day of -12.77%, a worst 21-day stretch of -40.0%, 51 days below -3σ and 11 below -5σ .

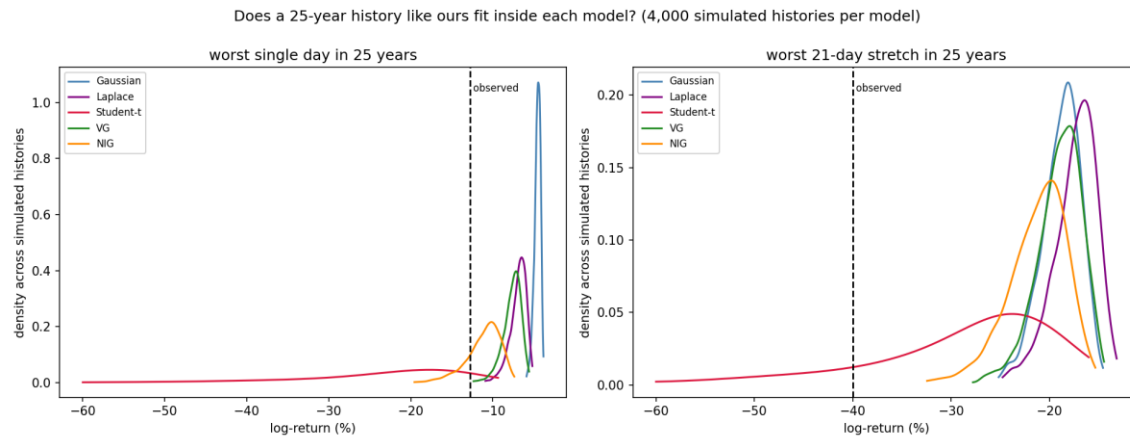


Figure 13. Distribution across 4,000 simulated histories of the worst single day (left) and worst 21-day stretch (right) per model, observed values dashed. Only the Student-t places real mass beyond both observed lines.

Three of the five models cannot generate this history at all: in 4,000 attempts the Gaussian's median worst day was -4.5% and it never came near -12.77%, the Laplace never reached it, the VG got there 14 times, and none of the three ever produced 11 days below -5σ .

The Student-t contains the observed worst day easily, in fact too easily: 91% of its histories hold a worse day and its median worst is -20.7%, beyond anything the index has done, the same shoulder-for-tail trade that failed the 95% backtest now visible as invented catastrophes.

The NIG is the best calibrated on single days, with the observed worst at its 19th percentile. But the worst 21-day stretch defeats everyone: history assembled -40% in a month, the t produces a month that bad in 17% of histories, the NIG in 0.03%, the rest never.

The timing deficit can be stated as a probability. Taking each model's 99% violation sequence from the Section 7 backtest, I held its total hit count fixed (forgiving every error of magnitude in advance), scattered the hits uniformly over the 5,787 days 20,000 times and recorded the largest number landing in any 21-day window.

Observed: 12 for the Gaussian and Laplace, 11 for the Student-t and NIG, all in late February 2020. Under independence the median largest cluster is 3 to 4 and the 99th percentile is 5 to 6; not one permutation in twenty thousand reached the observed value for any model. Even the models whose tails Section 8.1 vindicated found the timing of March 2020 essentially impossible.

8.4 The verdict

Table 11. The four crises sorted by both dimensions of surprise.

Crisis	Under the	Under the	Warning	Verdict
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	Gaussian	Student-t / NIG	beforehand	
Dot-com crash	Black (worst day once per 9,470 y)	Routine (1.5 to 2 y)	outside sample	Not a swan: a grind of ordinary days
GFC	Black (67 ages of the universe)	Rare but priced (5 to 40 y at the time)	visible build-up	Grey swan
COVID-19	Black (10^{22} y; 10^{40} at the time)	Once-in-a-generation at the time	none	Closest to a true Black Swan; magnitude priceable, timing not
Fed rate hikes	Marginal (28 y)	Routine (< 1 y)	n/a	White swan under every model

Taleb's accusation, that finance manufactures its own swans by assuming thin tails, is exactly what Table 9 shows. But the analysis also sharpens the accusation in a way a purely verbal argument cannot: after the best marginal has done its work, what remains outside every model's expectation is not how bad the crisis days were but their refusal to arrive independently.

The magnitude swan is curable by a better distribution. The timing swan is the volatility clustering of Cont (2001), and within an iid model class it is incurable by construction.

9. Conclusions

9.1 The scorecard

Table 12. The five models across the whole project. The VG's Bayesian and backtest columns are carried by the Laplace, its symmetric special case.

	Gaussian	Laplace	Student-t	VG	NIG
Parameters	2	2	3	4	4
Δ AIC vs Gaussian	0	-1,771	-1,803	-1,798	-1,851
KS test	rejected	borderline	borderline	passes	passes
99% ES (in-sample)	-3.24%	-3.91%	-5.81%	-4.28%	-5.19%
PPC verdict	fails beyond the centre	fails deep tail	tail quantiles pass; moments undefined	(via Laplace)	passes all marginal checks
Backtest 99% violations	152 ($\times 2.6$)	110 ($\times 1.9$)	104 ($\times 1.8$)	(via Laplace)	91 ($\times 1.6$)
Basel red-zone share	30.8%	18.7%	17.9%	(via Laplace)	14.1%
ES shortfall ratio (97.5%)	1.33	1.20	0.97	(via Laplace)	1.07
Worst COVID day: implied return period	5.6×10^{22} y	62,400 y	10.6 y	11,400 y	119 y
Independence test	fails	fails	fails	(via Laplace)	fails

9.2 What I found

The Gaussian assumption fails where it is most expensive. The fitted v of 2.648, with its whole posterior interval above the variance singularity, says the tails are heavy as a stable population property. The price of ignoring that is concentrated in the tail the regulation cares about: 79.5% of missing 99% ES in-sample, 2.6 times the nominal violations out of sample, 31% of two decades in the Basel red zone, and breach-day losses one third beyond the booked ES.

Most of the gain is exponential tails; the last part is the NIG's. The Laplace, at the Gaussian's own parameter count, captures 96% of the best model's AIC improvement, so the bulk of the fix is simply admitting that tails are not Gaussian. The NIG earns the remainder with real structure: the only KS pass worth the name, every posterior predictive check including skew and the FRTB ES, a posterior β negative across its entire

interval, and the closest-to-nominal coverage at every backtest level. Between them sits the anomaly I found most instructive: the Student-t, the classic heavy-tail fix, is the worst 95% forecaster in the table because its tail is bought from its shoulders.

The Black Swan is model-relative, and mostly a Gaussian artifact. The same day is four trillion ages of the universe away under the Gaussian and a bad decade under the Student-t, with within-model parameter uncertainty spanning tenths of a decade against 21 decades of disagreement between models. Two refinements came out of taking the question seriously.

Heavy tails are not interchangeable where it matters: the exponential-tailed Laplace and VG, near-optimal by likelihood, misprice the extreme days by three to four orders of magnitude. And robustness to the estimation window is its own axis of model quality: refitted on calm pre-crisis years, the Student-t kept its tail while the NIG slid towards its Gaussian limit and gave up three orders of magnitude of foresight, a fragility with direct consequences for any rolling-window internal model.

No static distribution passes the timing test. Every model failed the independence test at every level, in-sample as the posterior predictive ACF check and out-of-sample as clustered violations in 2008 and 2020, exactly as Cont (2001) documents and exactly as Christoffersen (1998) found for the static forecasts in his original application. Section 8.3 states the same failure as a single probability: granted its own violation count, every model finds the March 2020 cluster a sub-1-in-20,000 arrangement. What stays wrong after the marginal is fixed is the arrival pattern of the violations, and that is a property of dynamics, which an independent-increments model discards by construction.

9.3 Limitations

There are six caveats I want to state plainly. Everything rests on daily log-returns of one equity index, and both the skew findings (equity-specific per Cont, 2001) and the tail estimates (which lighten to $v = 3.4$ at weekly frequency) are specific to that choice.

The model class is iid by design; I set out to measure what a static marginal can and cannot do, and the independence failures are that design's honest boundary, quantified rather than escaped. Sub-sample NIG fits are sometimes only weakly identified, sitting on the Gaussian-limit ridge where individual parameters mean little even though quantile forecasts stay well behaved.

The backtest's 500-day window and 21-day refit cycle are conventional but not innocent, and the Basel traffic-light figures are descriptive; I did not implement the full regulatory apparatus around the multiplier.

The Bayesian and backtest columns represent the VG family by its Laplace special case; given that the full VG sits between the Laplace and the NIG on every in-sample measure this is unlikely to change any ranking, but it is a corner I cut knowingly. And the return

periods of Section 8 extrapolate parametric tails far beyond the data: the empirical recurrence column disciplines the comparison inside the sample, but nothing in 25 years of returns can verify a 119-year claim, let alone a 10^{22} -year one. I read those numbers as statements about the models, not about the market.

9.4 Closing remark

I set out to ask whether heavy-tailed Lévy models materially improve tail risk measurement under the Basel metrics.

They do, by amounts that would change a capital number: roughly a third more Expected Shortfall at the FRTB's own confidence level, and half the regulatory red-zone time out of sample.

But the full battery of tests also exposed what stays broken: the arrival times of the violations, which no independent-increments model can place. A natural continuation would put these Lévy distributions to work as the innovation law inside a conditional volatility model, asking whether the NIG's tail and a dynamic scale together can pass the one test nothing in this paper passed. That division of labour, the size of tail risk against its timing, measured on one dataset with one consistent toolkit, is what this project contributes.

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